Binh Minh Tran

Biomedical AI · Graph and multimodal learning

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Research interests: Computational RNA biology; graph and multimodal learning; spatial omics; computational oncology; AI for molecular design.

EDUCATION

2027-2031 (planned)

Doctor of Philosophy, Medical Science

Computational RNA Biology, John Curtin School of Medical Research,
The Australian National University, Canberra, Australia.

- PhD supervision and funding confirmed with Professor Eduardo Eyras; subject to formal PhD admission.

Feb 2023-Present

Bachelor of Advanced Computing (Honours)

Computational Data Science, The University of Sydney, Australia.

- Expected completion: December 2026. Cumulative WAM: 82/100 (Distinction).
- Thesis: *Multimodal Breast Cancer Survival Prediction with Hierarchical Niche-Interface Graph Learning.* Supervisor: Associate Professor Xiuying Wang.

2019-2022

High School Diploma, Specialised Physics Programme

HUS High School for Gifted Students, VNU University of Science,
Vietnam National University, Hanoi, Vietnam.

- GPA: 9.5/10. Outstanding academic achievement award, 2019-2022.

RESEARCH EXPERIENCE

Oct 2026-Present

Visiting Researcher

Centre for Computational Biomedical Sciences, John Curtin School of Medical Research, The Australian National University.

- Undertaking preliminary research in computational RNA biology with Professor Eduardo Eyras in preparation for doctoral study.
- AI/ML-based interpretation of genetic variants affecting RNA modifications (m6A, pseudouridine and m5C), distinguishing predicted modification changes from disease-relevant functional effects.

Feb-Oct 2026

Honours Research Student

School of Computer Science, The University of Sydney.
Supervisor: Associate Professor Xiuying Wang.

- Designed NIMSurv, a hierarchical graph model integrating clinical data and imaging mass cytometry for breast-cancer survival prediction, with interpretable contributions from tissue interfaces.
- On 544 METABRIC patients, achieved 10-year disease-specific survival Uno's C of 0.703 under patient-level 5 × 5-fold cross-validation (+0.078 versus a clinical-plus-summary baseline).
- Exploratory transfer of a frozen common-input model to Basel yielded 5-year Uno's C of 0.725. An earlier model, HiNT-Surv, is under review [M4].

RESEARCH EXPERIENCE - CONTINUED

Dec 2025-Present

Research Collaborator

Department of Obstetrics & Gynaecology, Hanoi Medical University, Vietnam.

- Developed and implemented a multimodal, multitask model integrating clinical variables, H&E whole-slide images and tumour RNA-seq to predict ER, PR and HER2 status in breast cancer [C1].
- Conducted patient-level stratified five-fold cross-validation on 937 TCGA-BRCA cases (macro AUC 0.890) and frozen external testing on 122 CPTAC-BRCA cases (macro AUC 0.931; no retraining, tuning or threshold adjustment).

Dec 2025-Mar 2026

Research Assistant

Center for Biomedical and Pharmaceutical Research,
Vietnam Academy of Science and Technology (VAST), Vietnam.

- Applied reproducible NGS preprocessing and annotation workflows, including FastQC, Trimmomatic, genome assembly, gene prediction and functional annotation, to mutation and microbial gene-function analyses.

Dec 2024-Dec 2025

Research Collaborator

Hanoi University of Industry, Vietnam.
Supervisor: Associate Professor Dang Hung Tran.

- Developed and implemented two graph-learning models for lncRNA-disease association prediction: a variational graph neural network with multi-view integration [C2] and a dual-stream contrastive graph model [C3].
- Evaluated both models and drafted the manuscripts, contributing to two peer-reviewed conference publications.

FUNDED RESEARCH PROJECT

Dec 2026-Nov 2027 (upcoming)

AI Model for Individualised Post-Test Probability after High-Risk NIPT

Department of Obstetrics & Gynaecology, Hanoi Medical University.

- Co-principal investigator with Nguyen Van An. Funding: VND 10,000,000 awarded by Hanoi Medical University. Supervisor: Nguyen Thi Bich Van, MD, PhD.
- Develop and calibrate an individualised probability model for confirmed trisomy 21, 18 or 13 following a high-risk NIPT result, using signed Z-score, fetal fraction and assay platform in retrospective high-risk referrals.

INDUSTRY EXPERIENCE

Dec 2024-Mar 2025

Data Engineer Intern

Viettel Software, Hanoi, Vietnam.

- Built and maintained Python/SQL ETL pipelines; contributed to data-quality monitoring, schema design and pipeline/query performance tuning.

Jun-Jul 2024

Data Engineer Intern

Agribank, Hanoi, Vietnam.

- Supported ETL into reporting systems and contributed to dashboards and operational reports for internal stakeholders.

PUBLICATIONS

Applicant's name in bold. † Equal contribution.

Published peer-reviewed papers

[C1] B. M. Tran†, V. A. Nguyen†, L. H. Nguyen, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, T. B. V. Nguyen, and D. H. Tran. "A Multimodal Deep Learning Framework for Breast Cancer Receptor Status Prediction." *IUKM 2026*, LNAI 16918, Part I, Springer, pp. 102-114, 2027. DOI | Full paper.

[C2] L. H. Nguyen, **B. M. Tran**, and D. H. Tran. "A Variational Graph Neural Network with Multi-view Integration for Predicting LncRNA-Disease Associations." *Recent Challenges in Intelligent Information and Database Systems (ACIIDS 2026)*, CCIS 2964, Springer, pp. 355-369, 2026. DOI | Full paper.

[C3] L. H. Nguyen, **B. M. Tran**, and D. H. Tran. "Dual-Stream Contrastive Graph Representation Learning for lncRNA-Disease Association Prediction." *Proceedings of the FISU Joint Conference on Artificial Intelligence (FJCAI 2026)*, pp. 151-158, 2026. ISBN 978-604-45-2586-0.

Manuscripts under review

[M1] B. M. Tran†, V. A. Nguyen†, Q. A. Dang, N. M. H. Nguyen, Q. A. Nguyen, H. N. Tran, D. H. Tran, and T. B. V. Nguyen. "Development and cross-cohort evaluation of a multimodal deep-learning model for ER, PR and HER2 status classification in breast cancer." Under review at *Scientific Reports*, 2026.

[M2] B. M. Tran, D. Q.-A. Nguyen, M.-H. Ha, D. M. Pham, T. D. Nguyen, Q. B. Doan, D. T. Doan, M.-A. Nguyen, D.-T. Nguyen, and O. T.-C. Chen. "H2F-GCN: Skeleton-based Hypergraph Frequency Coordinate Fusion Graph Convolution Network for Physical Rehabilitation Assessment." Under review at *IEEE Access*, 2026.

[M3] T. B. V. Nguyen, N. M. H. Nguyen, **B. M. Tran**, Q. A. Dang, and V. A. Nguyen. "A Deep Learning Model Supporting Classification and Localisation of Breast Tumour Lesions on Ultrasound." In Vietnamese. Under review at *Journal of Medical Research (Hanoi Medical University)*, 2026.

[M4] B. M. Tran, L. Zhang, and X. Wang. "HiNT-Surv: Hierarchical Niche-Aware Multi-Microenvironment Learning from Imaging Mass Cytometry for Survival Prediction." Under review for the *IEEE Symposium on Computational Intelligence in Health and Medicine (CIHM), SSCI 2027*.

SCHOLARSHIPS

2027 (prospective)

MACSYS PhD Scholarship

ARC Centre of Excellence for the Mathematical Analysis of Cellular Systems (MACSYS), The Australian National University.

- Supports doctoral research in Medical Science (Computational RNA Biology) at the Centre for Computational Biomedical Sciences, John Curtin School of Medical Research.
- Funding confirmed in writing by Professor Eduardo Eyras, subject to formal PhD admission.

2023-Present

Sydney International Student Award

The University of Sydney.

- Tuition support awarded at admission for the duration of the degree.

AWARDS

2026

Top 10 Finalist, AURORA Scientific Research Awards

Vietnam-Australia Innovation Network (NIC-AU).

- Grand Final scheduled for December 2026, Hanoi.

2026

Outstanding Student Award

Embassy of Vietnam in Australia.

- Academic and research achievement.

2026

Second Prize, Vietnamese Young Intellectuals Forum in Australia

Breast-cancer receptor-status research [C1].

2026

Third Prize, Youth Science and Technology Conference

Hanoi Medical University [C1].

- Selected for the 23rd Youth Science and Technology Conference of the Vietnamese Health Sector.

2025

January Star Award

Vietnam National Union of Students.

- Academic achievement, research and student leadership; one of 117 national recipients.

TECHNICAL SKILLS

Programming	Python, R, SQL, Bash; Git, Linux, LaTeX, experiment tracking.
Frameworks	PyTorch, PyTorch Geometric, scikit-learn.
Methods	Graph neural networks; multiple-instance, self-supervised and contrastive learning; multimodal fusion; Cox and discrete-time survival models.
Evaluation	Patient-level nested cross-validation; bootstrap confidence intervals; paired tests with multiple-testing correction; effect sizes; calibration; external validation.
Biomedical data	Imaging mass cytometry; RNA-seq and gene expression; H&E whole-slide images; NGS preprocessing and annotation; lncRNA-disease networks.
Languages	Vietnamese (native); English (IELTS 7.0); French (DELF A1).